

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1011

Wixom 138 kV
April 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1011. The Interconnection Customer proposes a 53.3 MW solar photovoltaic facility interconnecting at the Wixom 138 kV substation in the Kentucky area. The request was received on June 06, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 297 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,432,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Valemount / Tristate Substation 345 kV Line
Contingency: Sporn / Gavin Power Llc Substation 230 kV
Loading: 122.2% (Rating: 1,168 MVA)
DFAX: 9.1%
Pre-project loading: 121.8%
2. Facility: Pence / Valemount 345 kV
Contingency: Sporn / Gavin Power Llc 230 kV Line
Loading: 111.3% (Rating: 1,069 MVA)
DFAX: 8.7%
Pre-project loading: 110.8%
3. Facility: Henry Creek / Ubc S Campus 138 kV
Contingency: Albion Rd Substation / Platte City 138 kV L
Loading: 108.0% (Rating: 182 MVA)
DFAX: 39.9%
Pre-project loading: 96.3%
4. Facility: Toledo Bend Substation / Wharton 230 kV
Contingency: Barrett / Norwich 500 kV
Loading: 101.1% (Rating: 702 MVA)
DFAX: 6.9%
Pre-project loading: 100.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Valemount / Tristate Substation 345 kV Line	Rebuild line with higher-capacity conductor	\$118,200,000	1% (\$753,000)
Pence / Valemount 345 kV	Rebuild line with higher-capacity conductor	\$58,297,000	1% (\$384,000)
Henry Creek / Ubc S Campus 138 kV	Reconductor line	\$2,677,000	9% (\$232,000)
Toledo Bend / Wharton 230 kV	Reconductor line	\$5,662,000	1% (\$63,000)
	Total allocated to project		\$1,432,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.